Design Guideline: Interior Lighting Electronics

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Scope

This guideline applies to the electronics of the Interior Lighting system of the Nova EV: the Ambient Light Controller (CMP-ILT-01), the LED Light Strip (CMP-ILT-02), the Reading Lamp (CMP-ILT-03), the Footwell Light (CMP-ILT-04), the Dome Light (CMP-ILT-05), the LED Driver Board (CMP-ILT-06), the Light Guide (CMP-ILT-07) and the Touch Switch (CMP-ILT-08).

Rules

DG-ILT-01 Current limit per LED channel

Each LED channel shall be current-limited to 150 mA by hardware on the LED Driver Board, independent of software. A software failure shall never lead to an LED current above the limit. REQ-ILT-0001 (Reading Lamp: Limit current per channel) implements this rule.

DG-ILT-02 Dimming with vehicle speed

Ambient light intensity shall be reduced linearly from 100 percent at standstill to 40 percent at 130 km/h. The dimming curve shall be a calibration table. REQ-ILT-0004 (LED Light Strip: Dim ambient light with vehicle speed) implements this rule.

DG-ILT-03 Switch-off after timeout

Any interior light that is switched on manually shall switch off automatically after a configurable timeout, 10 minutes by default, when the vehicle is locked. REQ-ILT-0005 (Footwell Light: Switch off after a timeout) and REQ-ILT-0090 (Touch Switch: Switch off after a timeout) implement this rule. REQ-ILT-0090 has no test set yet and is reported under the Requirement Traceability Policy, rule RTP-01.

DG-ILT-04 Colour consistency

All ambient light sources in the cabin shall show the same colour within a tolerance of 0.01 in the CIE 1931 colour space when set to the same drive mode. REQ-ILT-0059 (Ambient Light Controller: Match light color to the drive mode) and REQ-ILT-0062 (LED Light Strip: Match light color to the drive mode) implement this rule.

DG-ILT-05 Fade times

Switching lights on or off shall use a fade of 300 ms to 800 ms. Hard switching is allowed only for the hazard warning flash function.

DG-ILT-06 LED failure reporting

The LED Driver Board shall detect an open or shorted LED channel and report it as a diagnostic error code. The remaining channels shall stay operational.

DG-ILT-07 Touch switch debouncing

The Touch Switch shall debounce inputs over at least 40 ms and shall ignore touches shorter than 30 ms, so that vibration does not trigger the switch.

DG-ILT-08 Communication with infotainment

The Ambient Light Controller shall receive the theme from the Head Unit (CMP-INF-01) over the vehicle network and shall keep the last theme when the connection is lost. REQ-ILT-0003 (Dome Light: Sync with the infotainment theme) implements this rule.

Verification

Rules DG-ILT-01 and DG-ILT-06 are verified with the test case types "Overvoltage 16 V" and "Fault injection". Rule DG-ILT-02 is verified with "Boundary values". Defect DEF-0375 (Ambient Light Controller: value out of tolerance), open with severity Critical on a test set required for Prototype Build Release (MS-C), is a violation of DG-ILT-04.

Applicability matrix

Rule	Components	Requirements
DG-ILT-01	LED Driver Board, Reading Lamp	REQ-ILT-0001
DG-ILT-02	Ambient Light Controller, LED Light Strip	REQ-ILT-0004
DG-ILT-03	Footwell Light, Touch Switch	REQ-ILT-0005, REQ-ILT-0090
DG-ILT-04	Ambient Light Controller, LED Light Strip	REQ-ILT-0059, REQ-ILT-0062
DG-ILT-08	Ambient Light Controller, Dome Light	REQ-ILT-0003